

Felipe Vasconcelos

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Education

University of Sao Paulo (USP) – MS in Data Engineering and Data Science

Expected June 2026

Federal University of Alagoas (UFAL) – BS in Computer Science

April 2024

Skills

Data and Machine Learning: Python, SQL, Pandas, Scikit-learn, NLP, LLMs, Prompt Engineering, Data Mining

Data Engineering and Big Data: Apache Spark, Apache Airflow, Delta Lake, PostgreSQL, Data Warehousing, ETL/ELT Pipelines, Vector Databases

Development and Tools: RESTful APIs, Apache Superset, Git, Linux, Jira, Scrum

Experience

Applied Researcher Engineer, Vonex – Remote, Brazil

Aug 2024 – Dec 2024

- Developed and deployed functional prototypes by integrating voice recognition AI models with vector databases for efficient data retrieval and analysis
- Evaluated and optimized the performance of various Large Language Models (LLMs) by applying systematic prompt engineering and data processing techniques
- Managed the end-to-end development lifecycle for new AI solutions, from conducting state-of-the-art research and experimentation to final prototype deployment
- Conducted applied research in machine learning, running targeted experiments to validate model effectiveness before developing and deploying solutions

Data Analyst, NEES – Remote, Brazil

Sep 2023 – Aug 2024

- Led the end-to-end creation of interactive dashboards using Apache Superset, from SQL querying and data processing to final publication, enhancing internal monitoring and strategic decision-making
- Developed RESTful APIs with Python and the Django framework to power a web application, enabling usability testing for a new AI solution across multiple user groups
- Architected and integrated dynamic data visualizations into a WordPress environment by authoring complex SQL queries, improving data accessibility for internal stakeholders
- Authored comprehensive technical documentation and provided direct user support to create custom reports and data visualizations based on stakeholder requirements

Research Scientist, UFAL – Maceio, Brazil

Aug 2021 – Aug 2024

- Designed and implemented scalable data processing and storage pipelines for Smart City projects using Apache Spark and Apache Airflow, enabling large-scale analysis of urban data
- Authored and published multiple research papers in recognized venues, including SBBD and JIDM, receiving the Best Paper award at the DSW-SBBD23 workshop for novel dataset contributions
- Engineered and released two public datasets on electoral and geospatial transportation data by managing the end-to-end data lifecycle from collection and cleaning to final publication
- Developed and proposed a novel framework for storing and analyzing Big Geospatial Data in a Data Lakehouse architecture, published as a recommended solution for Smart City environments

Projects

Data Lakehouses for geospatial data in smart cities

github.com/felipeVsc/lakehouses

- Architected and implemented an end-to-end Data Lakehouse using Delta Lake and a Medallion architecture to process and analyze real-time geospatial data from four major Brazilian cities.

- Developed automated data ingestion pipelines with Apache Airflow to collect live bus data from multiple city APIs, using Apache Spark for subsequent ETL processing and standardization.
- Authored and published the project's framework and findings in a peer-reviewed paper for the Brazilian Symposium on Databases (SBB D 2024), demonstrating its viability for smart city applications.

Event Data Warehouse for Storm Events Database

github.com/felipeVsc/stormevents

- Designed and implemented a Python framework with custom operators for machine learning and similarity searching.
- Developed a full ETL pipeline for the NOAA Storm Event Database, modeling the raw data into a star schema data warehouse to support efficient analytics.
- Applied the framework to generate key insights, successfully identifying geographic tornado clusters and enriching the dataset by classifying storm intensity with machine learning.

Textual data mining for categorizing news articles into topics

github.com/felipeVsc/textual_mining

- Built and trained a multi-class text classification model to categorize news articles into 42 topics, achieving an accuracy score that surpassed the 38% project baseline.
- Integrated generative NLP features to automatically produce impactful headlines and concise summaries, validated using text similarity metrics against the original articles.
- Engineered an end-to-end solution using a 210,000-article dataset to provide a practical tool for journalistic workflows.